



Lab Number & Title:

Lab 01 - [GitHub Identity, Student Portal Registration, and Ubuntu Server Installation]

Full Name:

Afifa Arooj

Registration Number:

FA24B1-SE-003

Course Code:

SE-665 – Cloud Computing

Section:

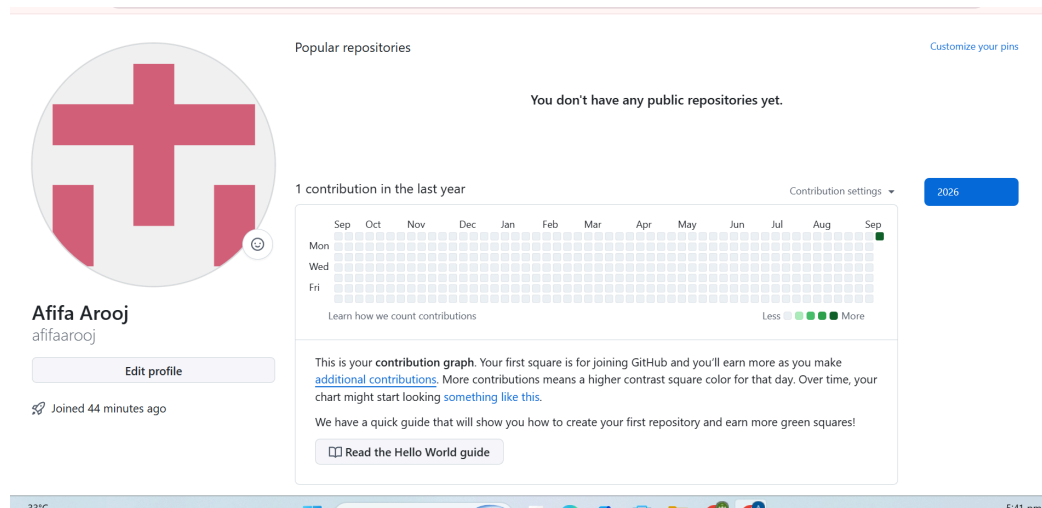
Section A

GitHub Username:

afifaarooj

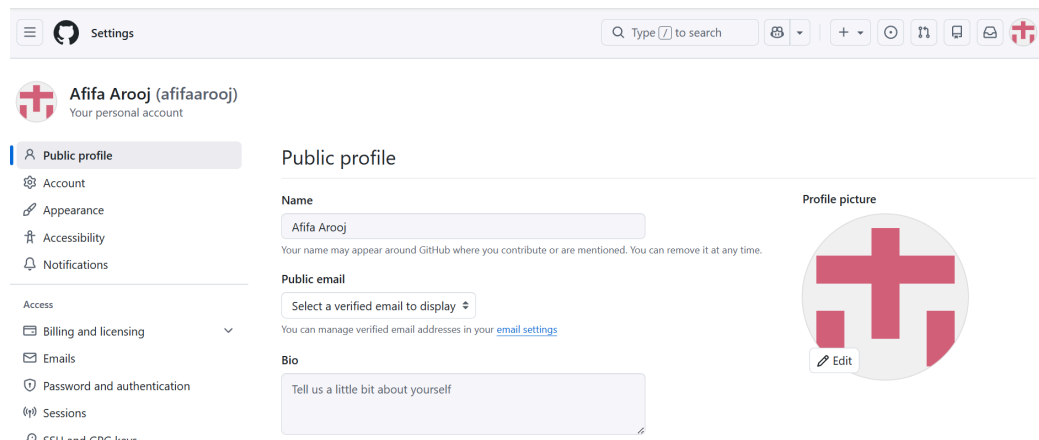
Task 1: Create and Configure Your GitHub Account

github_profile.png.



GitHub user profile page demonstrating account creation and username format.

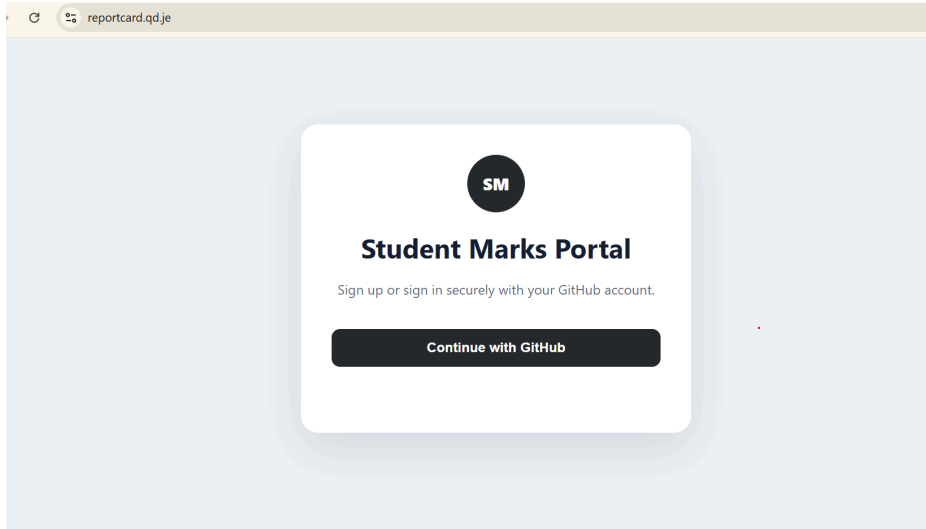
github_actual_name.png



GitHub Public Profile settings displaying the complete actual name field.

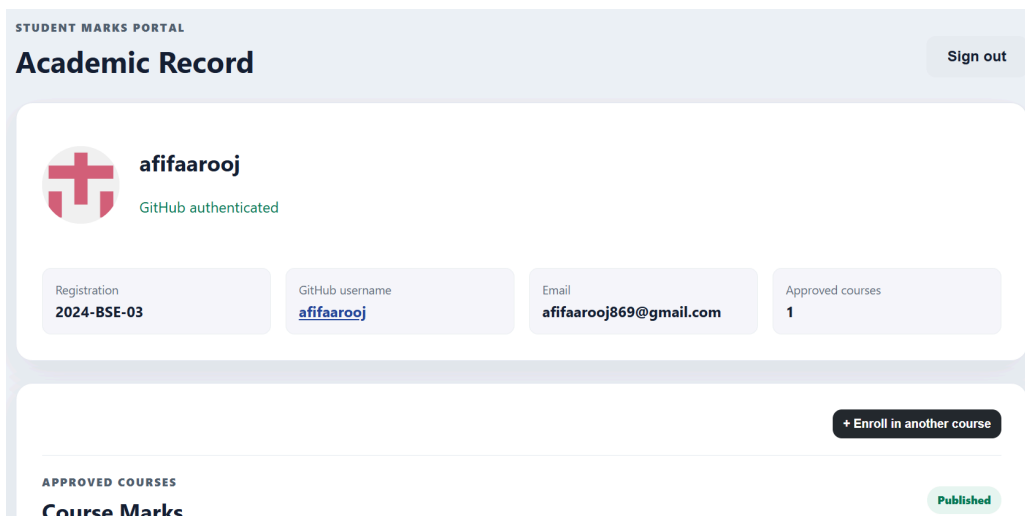
Task 2: Register in the Student Marks Portal

portal_login_page.png



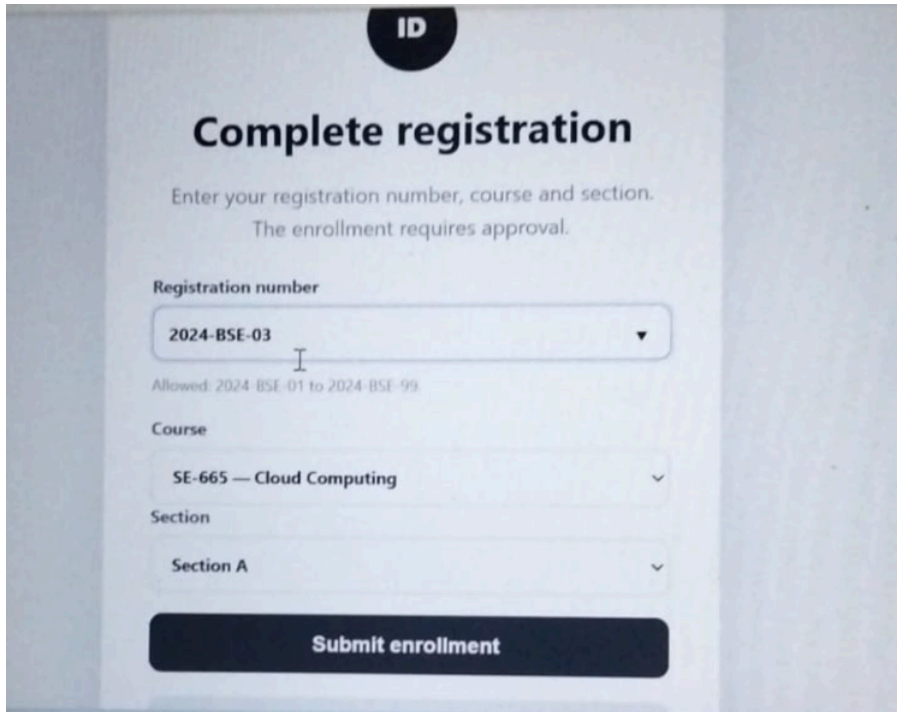
Main login interface for the Student Marks Portal.

portal_github_profile.png



Student Marks Portal showing authenticated GitHub account details.

portal_enrollment_submitted.png



Complete registration

Enter your registration number, course and section.
The enrollment requires approval.

Registration number

2024-BSE-03

Allowed: 2024-BSE-01 to 2024-BSE-99

Course

SE-665 — Cloud Computing

Section

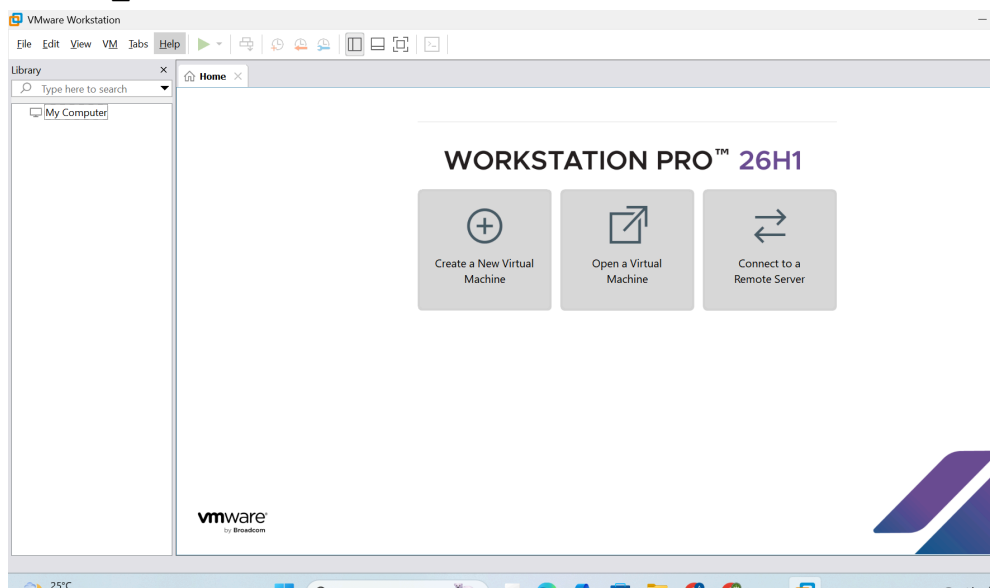
Section A

Submit enrollment

Submitted enrollment details showing registration number, course code, and section.

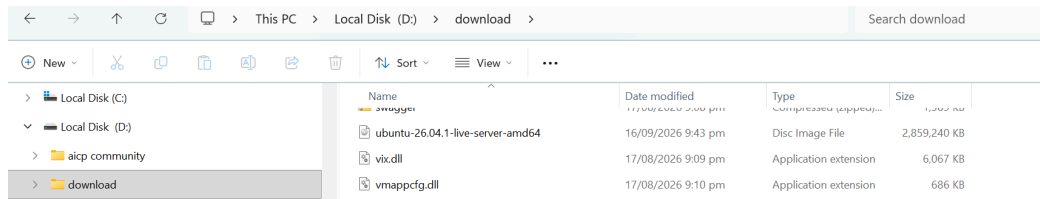
Task 3: Download VMware Workstation Pro and Ubuntu Server

vmware_workstation.png



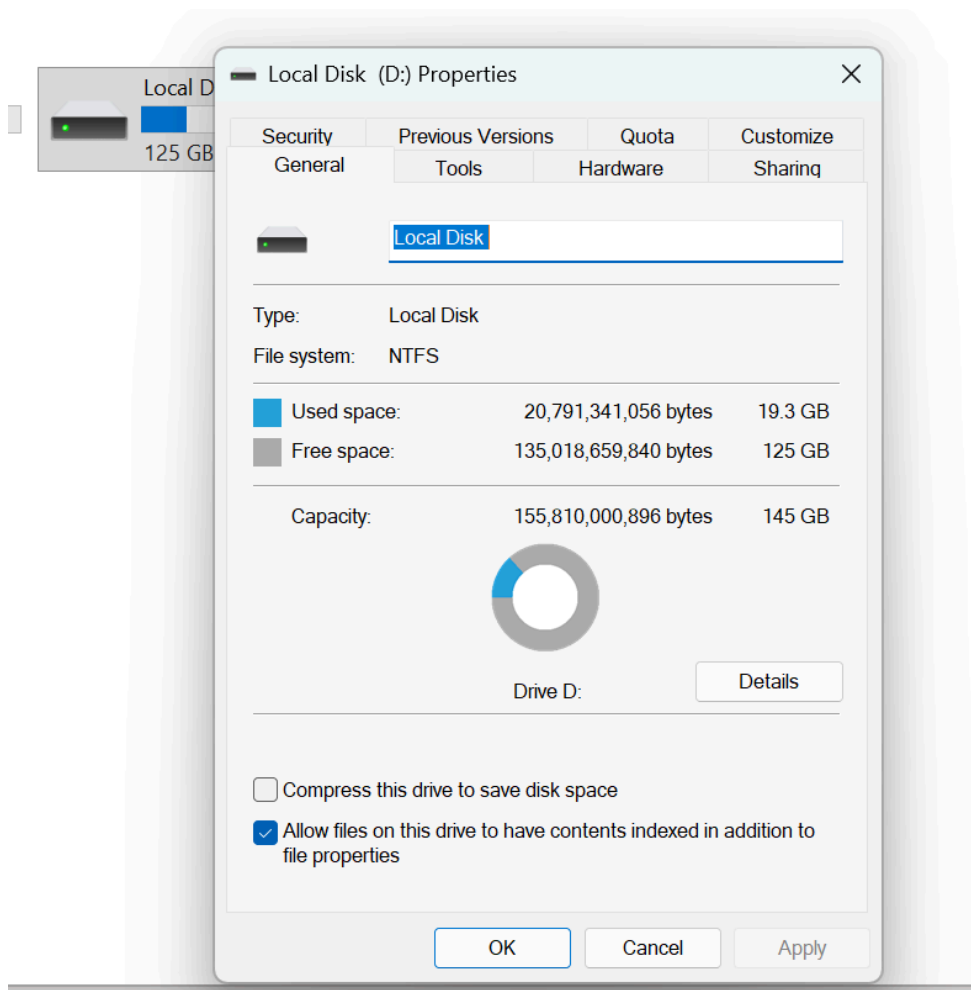
VMware Workstation Pro application running on the host system.

ubuntu_server_iso.png



Downloaded Ubuntu Server LTS ISO file stored locally.

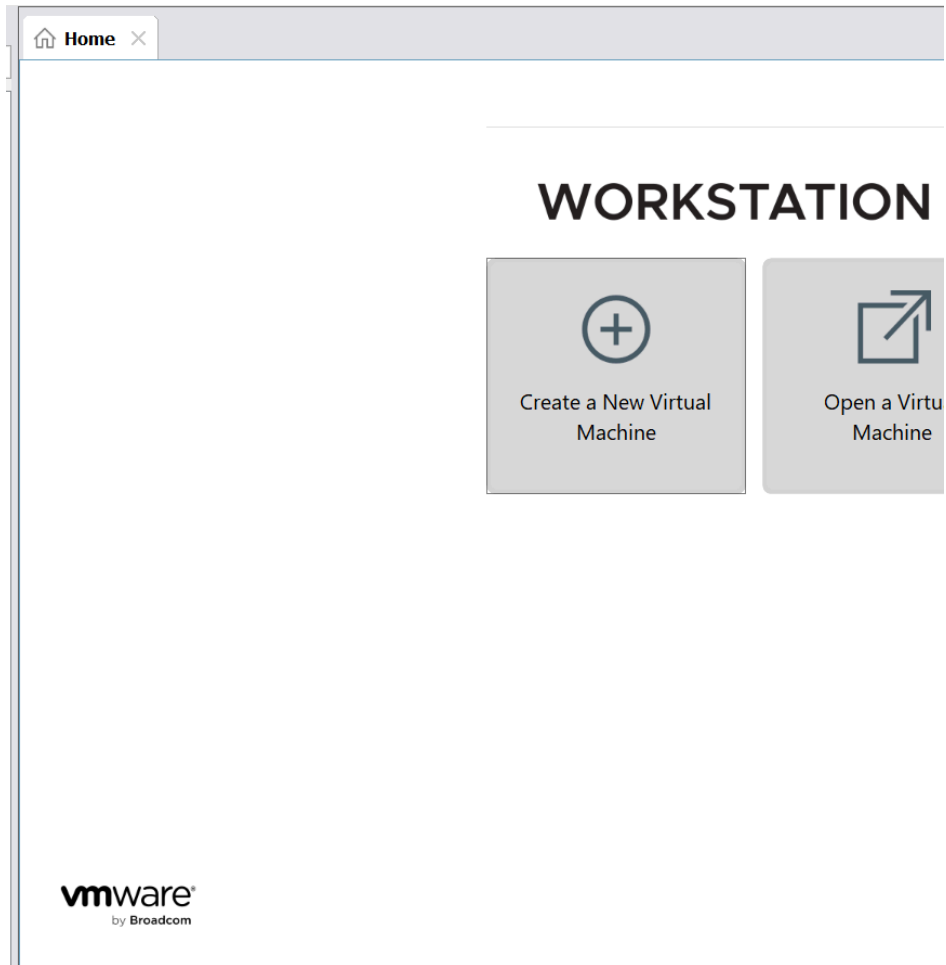
available_storage.png



Host disk properties confirming at least 20 GB of free storage space.

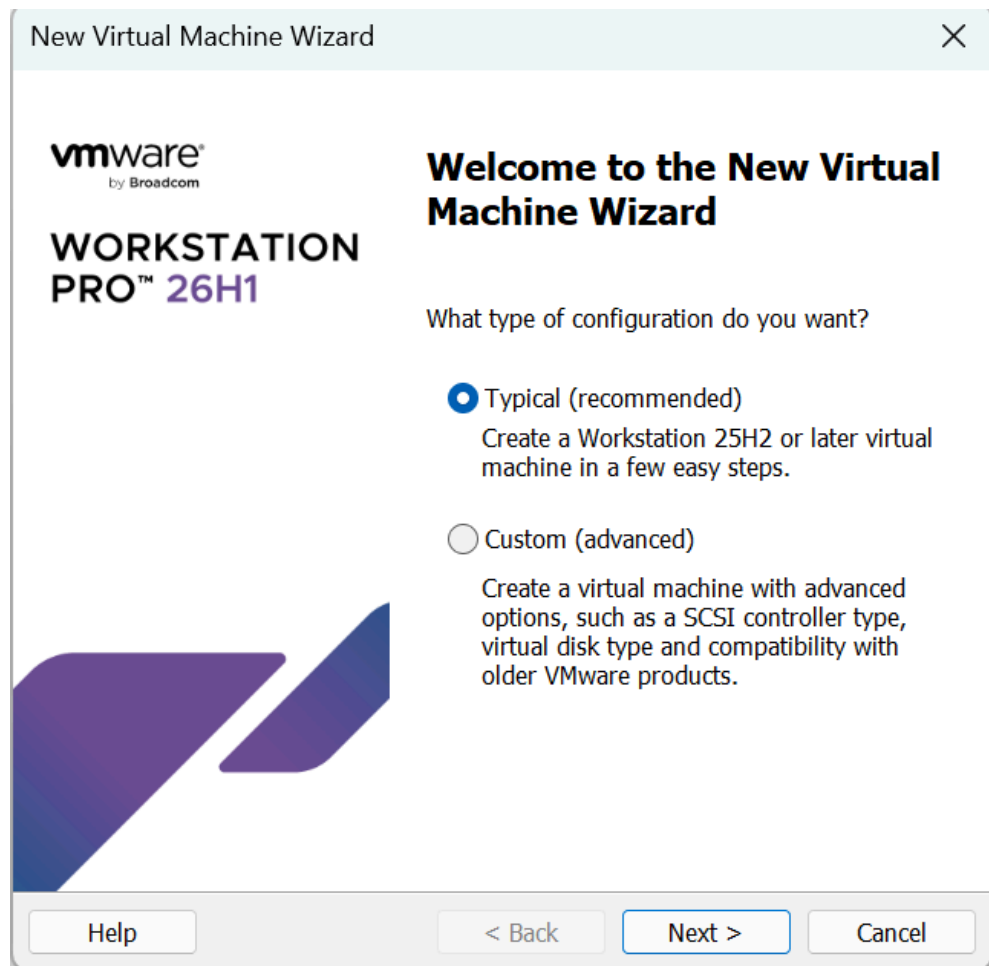
Task 4: Create the Ubuntu Server Virtual Machine

vm_creation_wizard.png



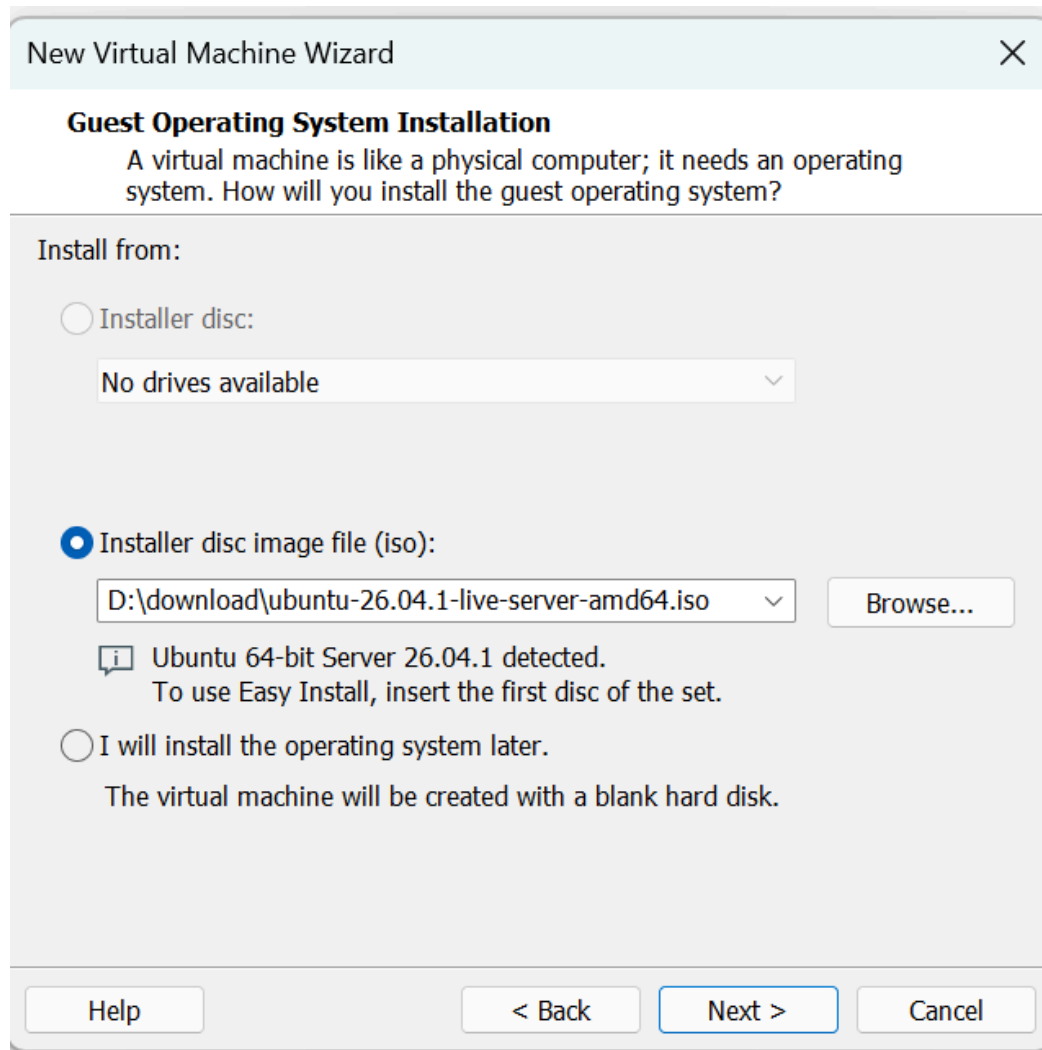
VMware New Virtual Machine Wizard initial screen.

vm_typical_configuration.png



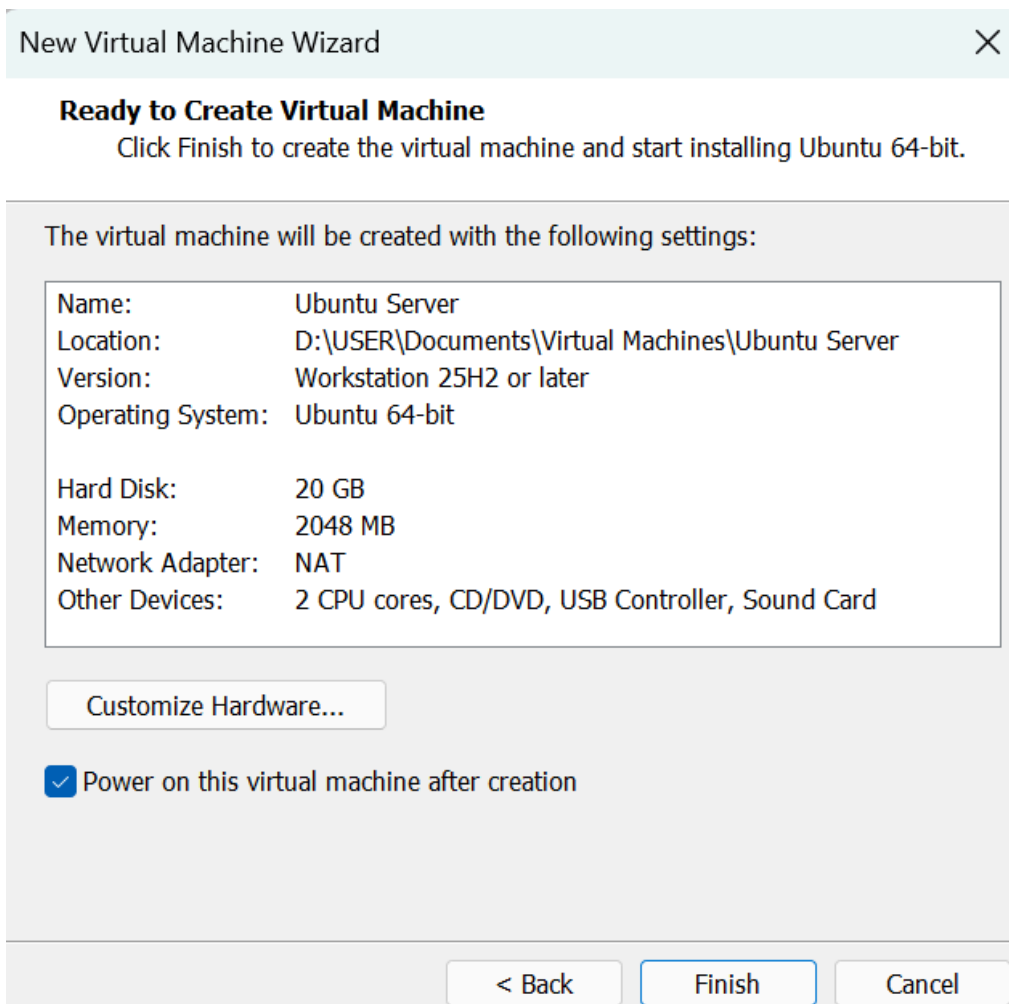
Selection of Typical virtual machine configuration.

vm_ubuntu_iso_selected.png



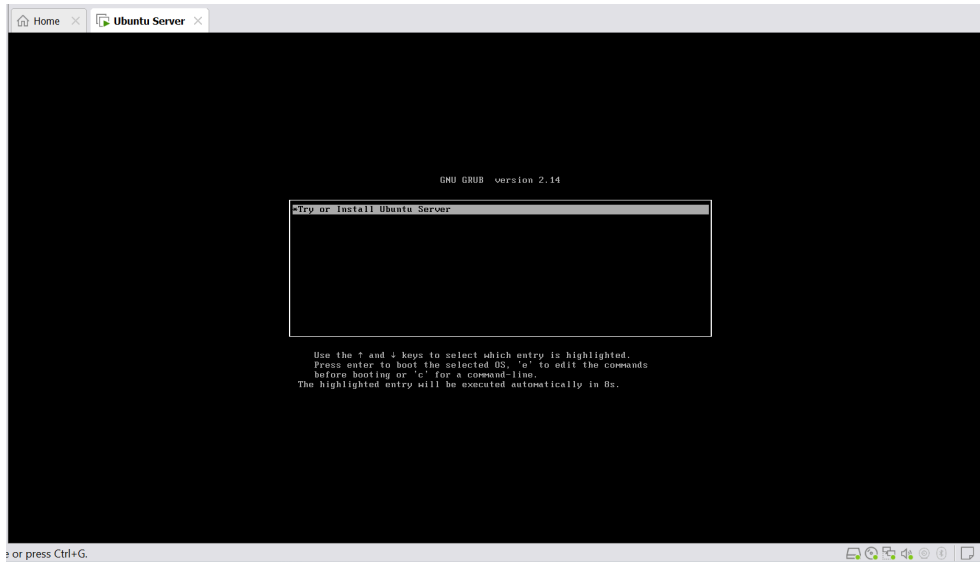
Target file path set to the downloaded Ubuntu Server ISO.

vm_configuration_summary.png



Virtual machine hardware specification summary before creation.

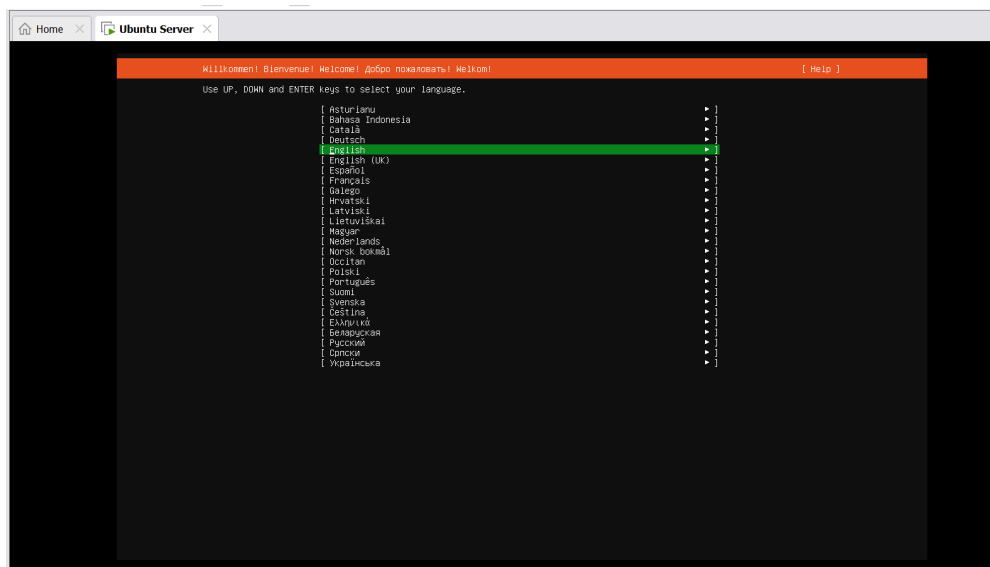
ubuntu_installer_boot.png



Initial boot screen of the Ubuntu Server installer inside VMware.

Task 5: Choose the Installation Language

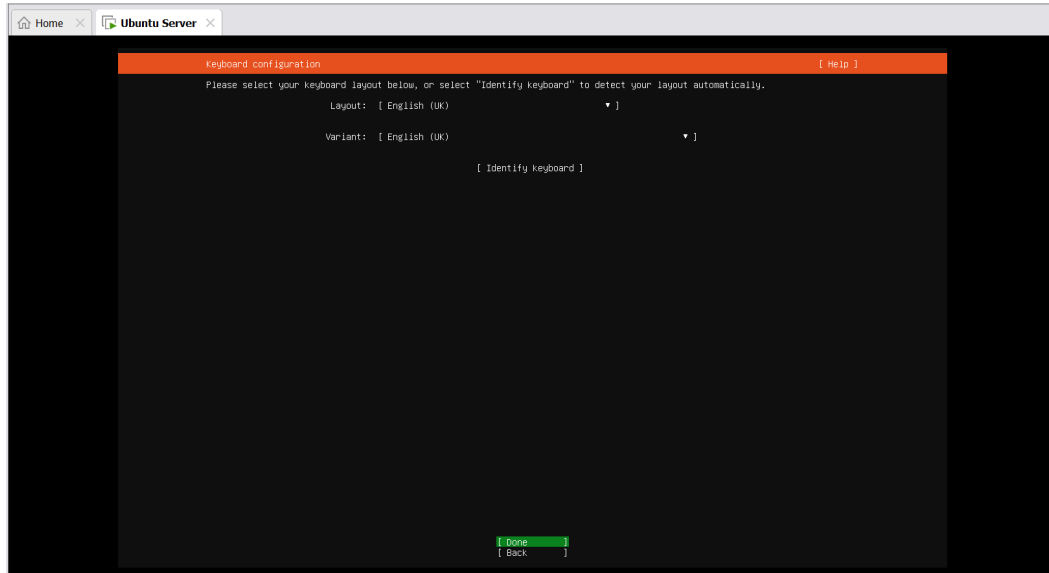
ubuntu_language.png



Installer language selection screen with English highlighted.

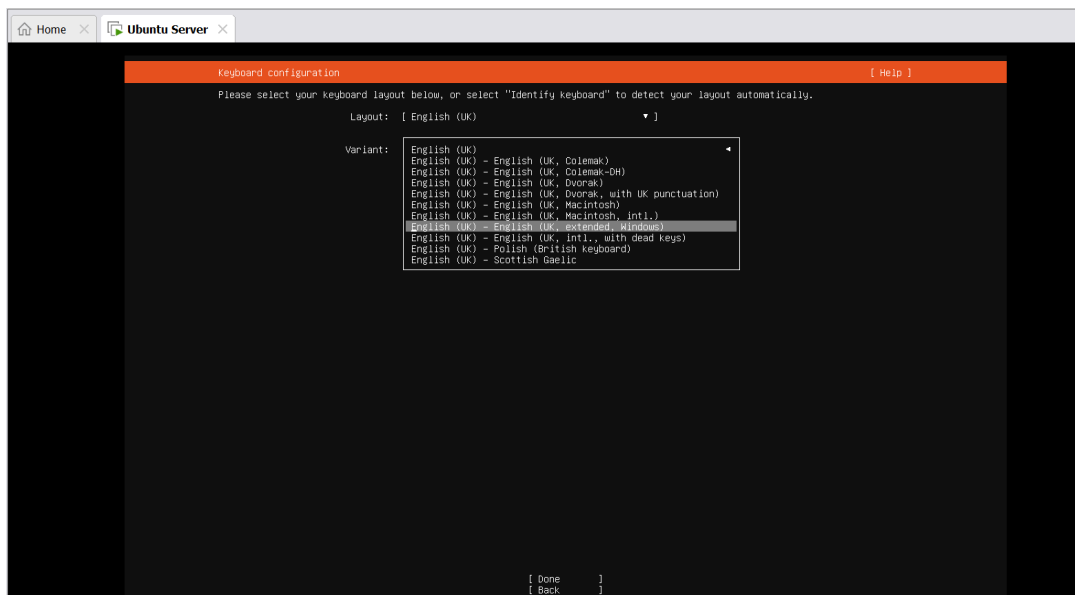
Task 6: Configure the Keyboard Layout

ubuntu_keyboard_layout.png



Keyboard layout selection prompt in the installer.

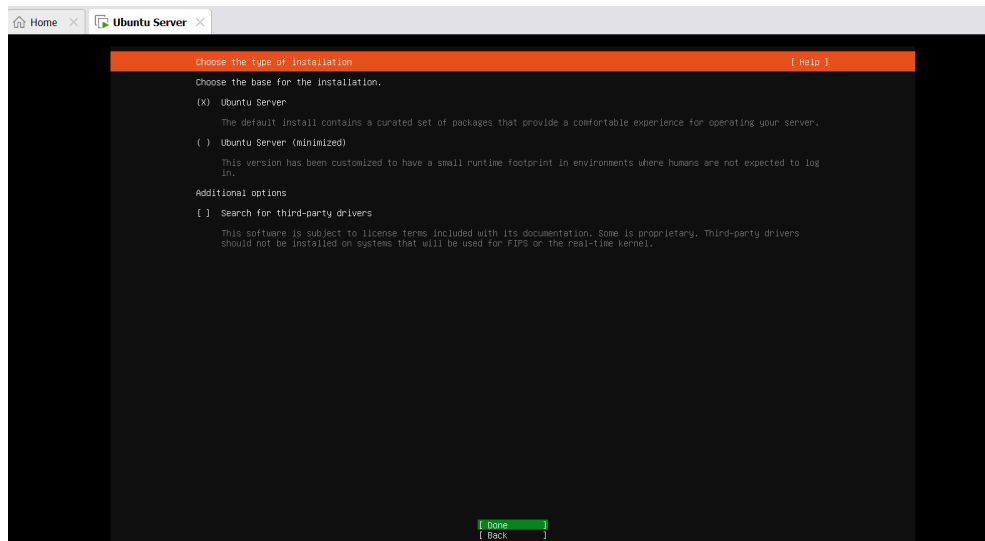
ubuntu_keyboard_variant.png



Selected keyboard variant configuration screen.

Task 7: Choose the Ubuntu Server Installation Type

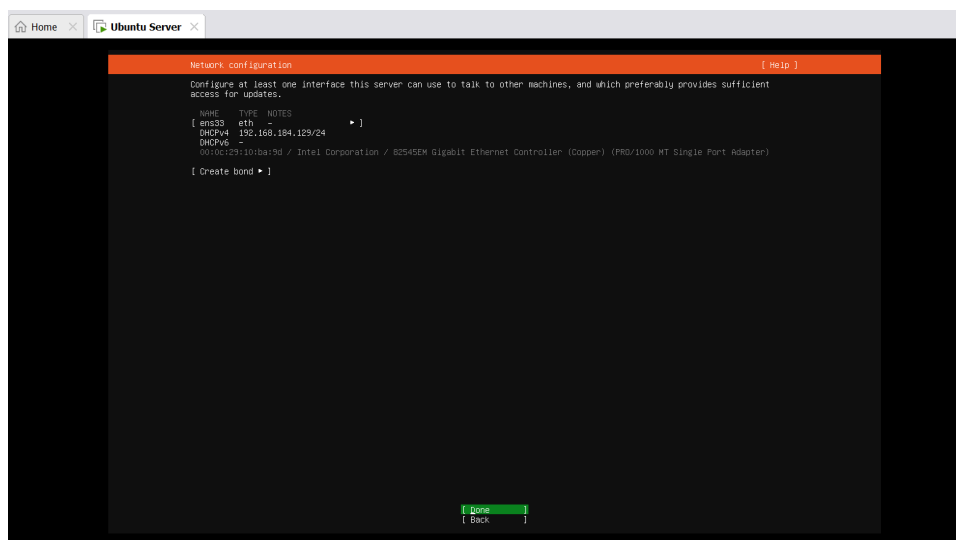
ubuntu_installation_type.png



Selection of the standard base Ubuntu Server installation type.

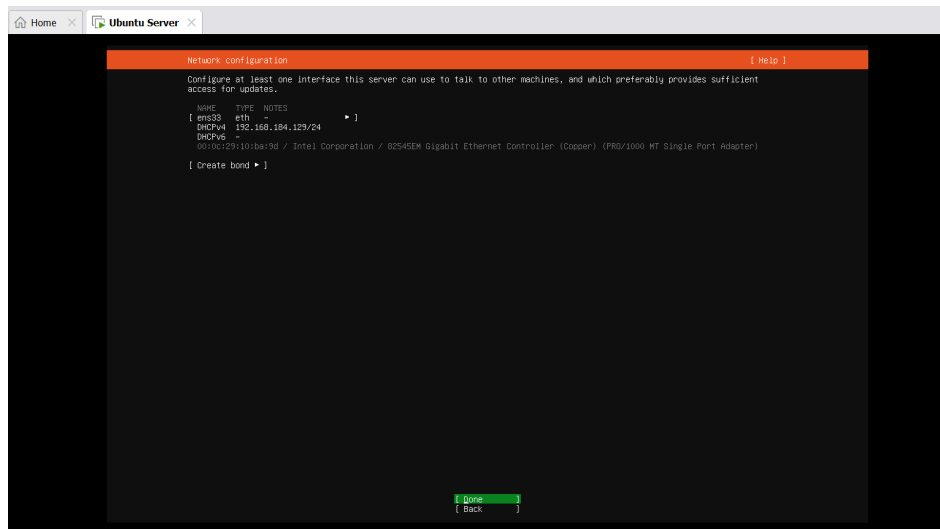
Task 8: Configure the Network

ubuntu_network_interface.png



Detected network interface listing in the installer.

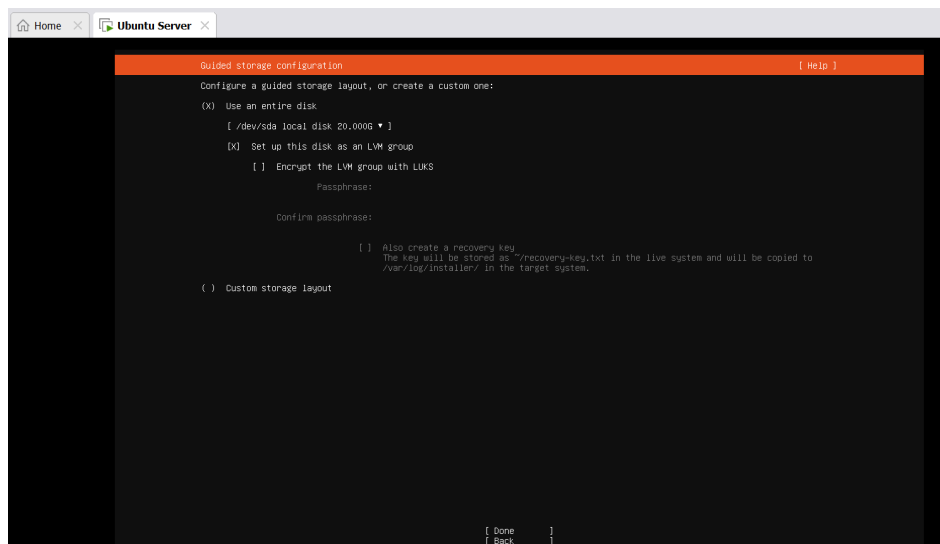
ubuntu_installer_network.png



Automatic IPv4 address assignment confirmed via DHCP.

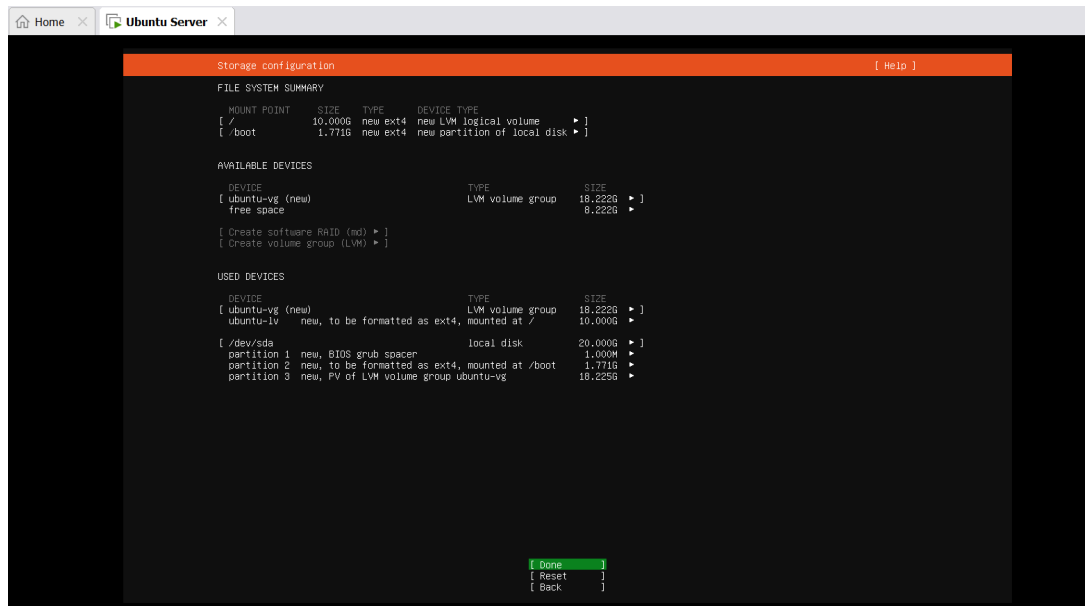
Task 9: Configure Storage

ubuntu_guided_storage.png



Guided storage option selected targeting the VMware virtual disk.

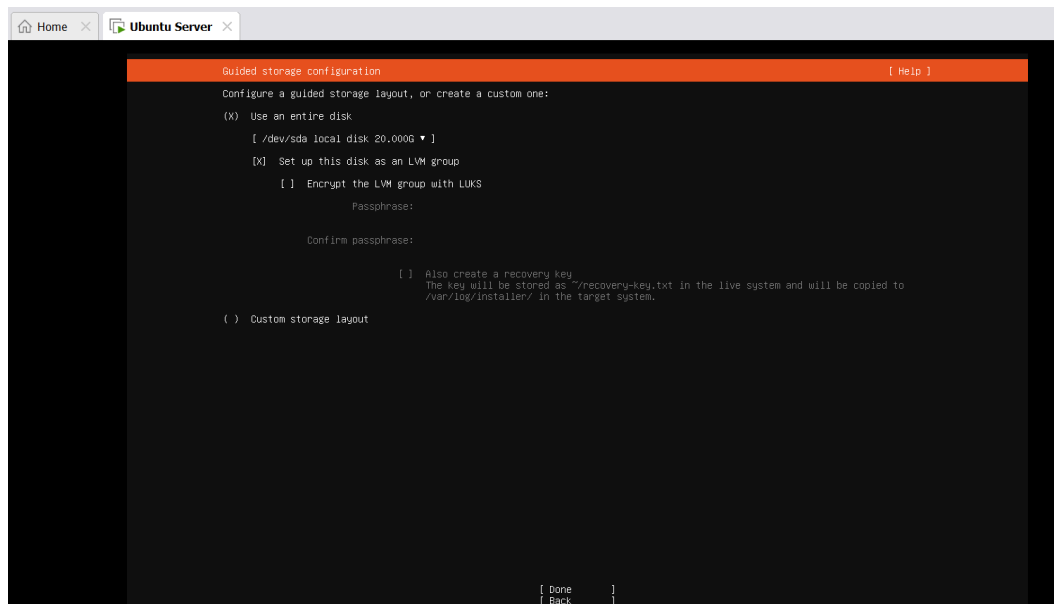
ubuntu_storage_configuration.png



Summary review of guided storage setup prior to disk allocation.

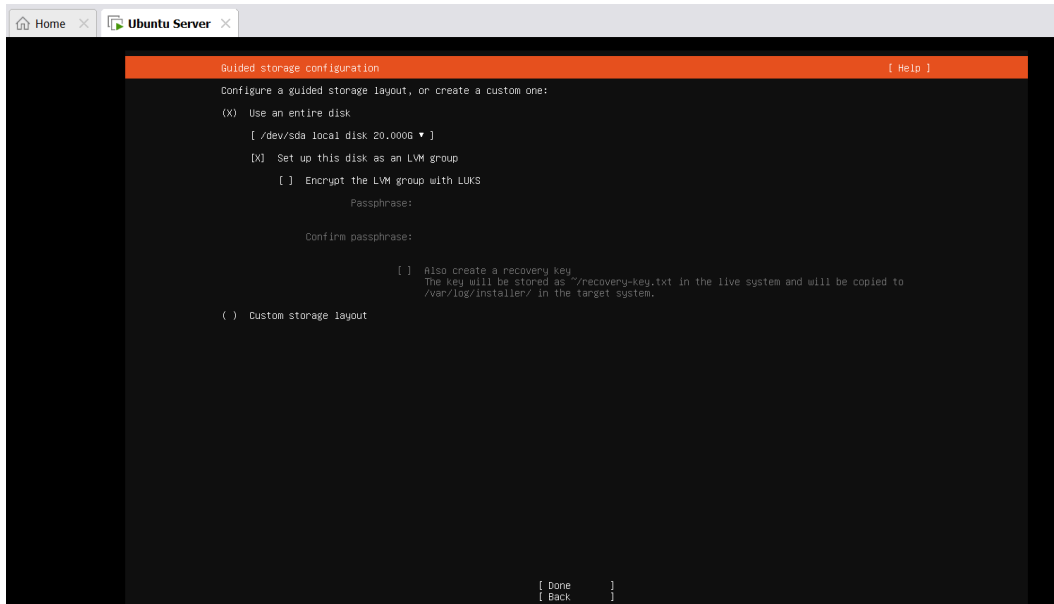
Task 10: Select the Installation Disk

Ubuntu_selected_disk.png



Target virtual disk selected for Ubuntu Server installation.

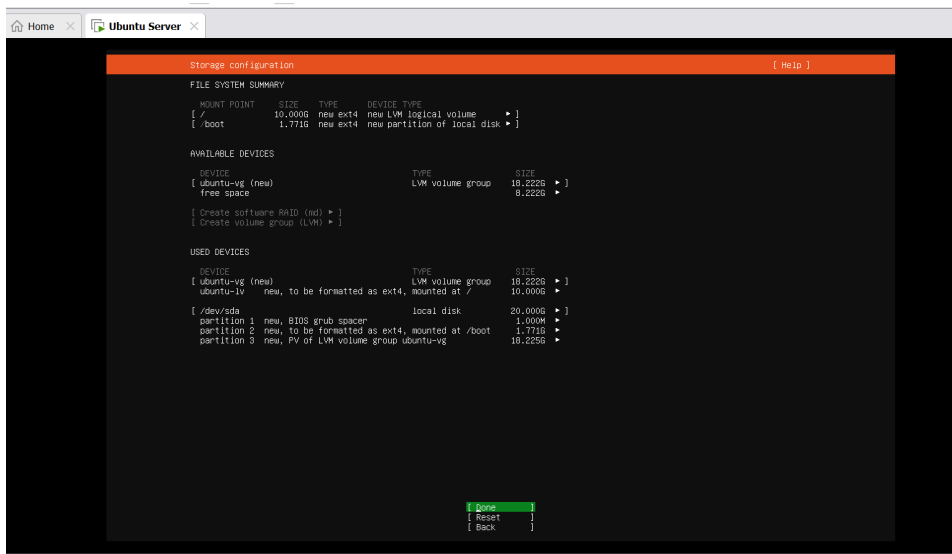
ubuntu_disk_details.png



Virtual disk capacity and name details verification.

Task 11: Review the Partitions

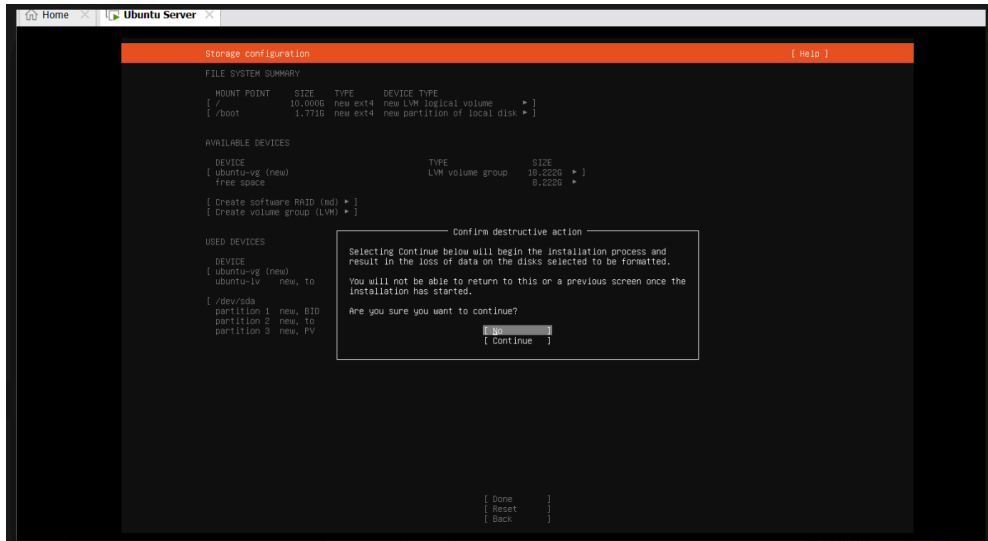
ubuntu_partition_layout.png



Proposed partition scheme review for the VMware virtual disk.

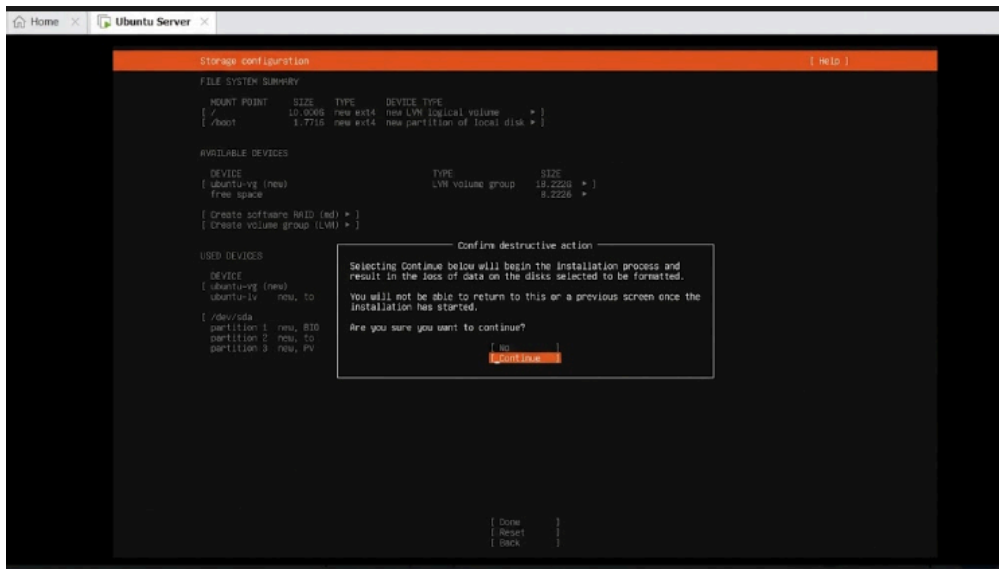
Task 12: Confirm the Storage Changes

ubuntu_storage_warning.png



Destructive action warning prompt displayed by the installer.

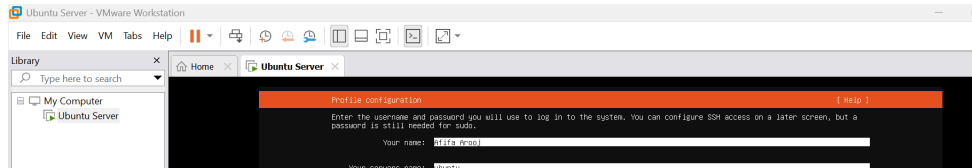
ubuntu_storage_confirmed.png



Final confirmation accepting partition table changes to the disk.

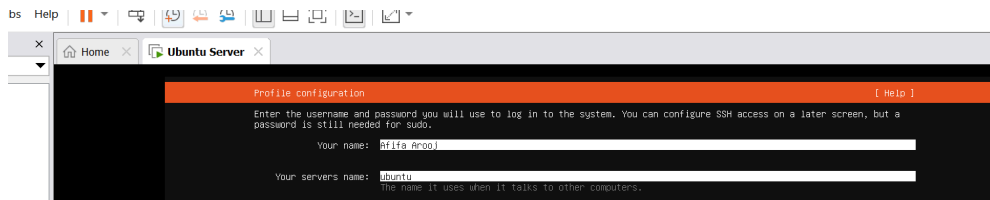
Task 13: Configure the Ubuntu Profile

ubuntu_actual_name.png



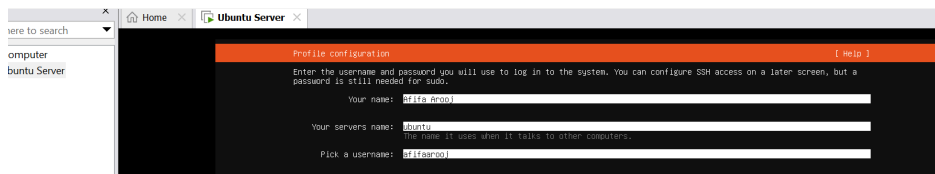
Profile setup with complete actual name entered in 'Your name' field.

ubuntu_server_name.png



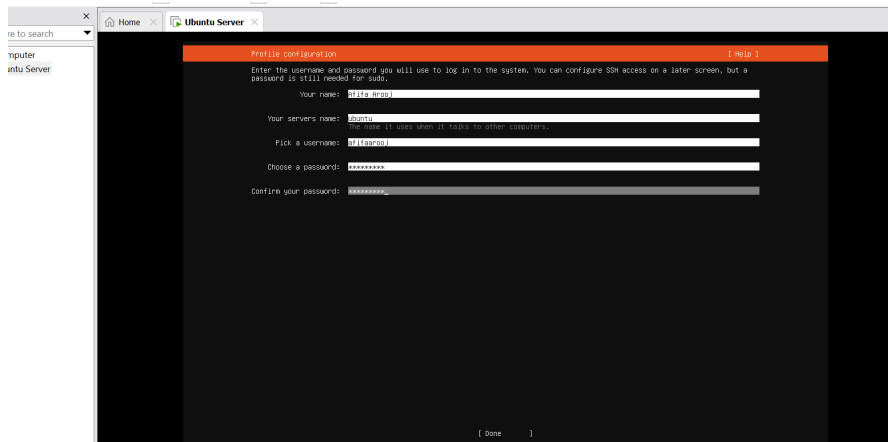
Hostname configured as 'ubuntu' in 'Your server's name' field.

ubuntu_username.png



System username matching exact GitHub username entered in 'Pick a username' field.

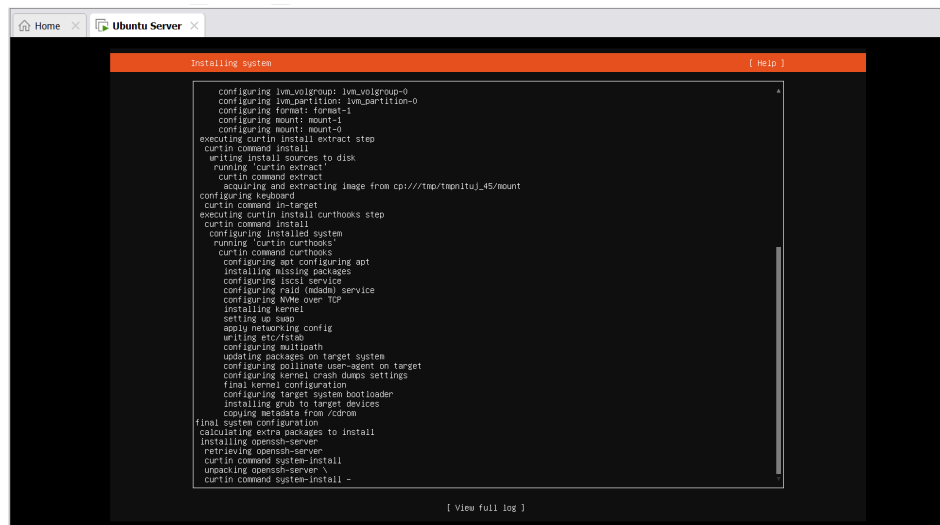
ubuntu_profile_setup.png



Completed identity profile form ready for submission.

Task 14: Complete the Software Installation

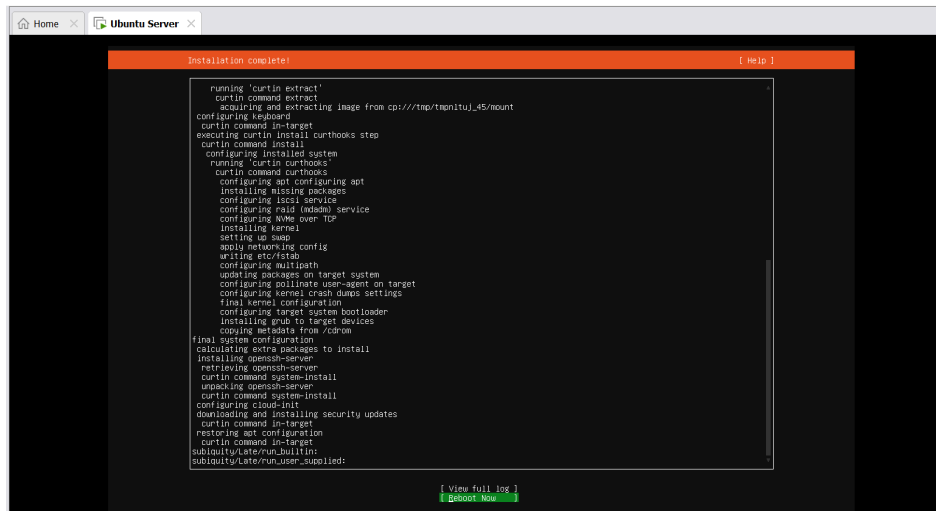
ubuntu_installation_progress.png



Base system installation and package extraction progress screen.

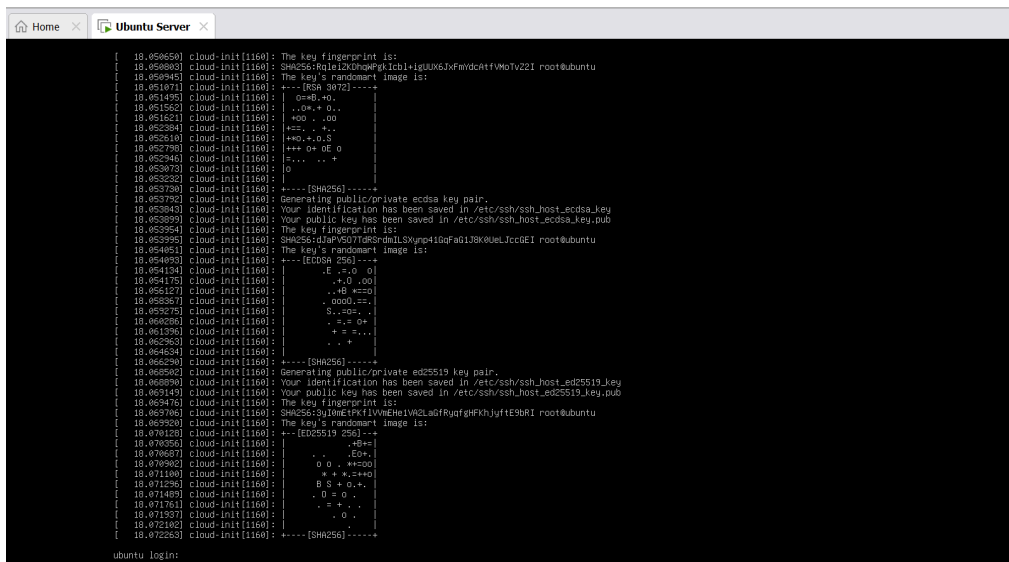
Task 15: Reboot and Verify the Ubuntu Identity

Ubuntu_installation_complete.png



Installation complete status with 'Reboot Now' prompt.

ubuntu_login_screen.png



System boot complete screen displaying Ubuntu login prompt.

ubuntu_terminal_login.png

Successful user authentication and initial shell session launch.

Ubuntu_identity_verified.png

```
Home x Ubuntu Server x
[ 18.468890] cloud-init[1160]: Your identification has been saved in /etc/ssh/ssh_host_ed25519_key
[ 18.469149] cloud-init[1160]: Your public key has been saved in /etc/ssh/ssh_host_ed25519_key.pub
[ 18.469476] cloud-init[1160]: The key fingerprint is:
[ 18.469706] cloud-init[1160]: SHA256:3y10nE1PKf1VnEHe1VA2La6fRyGfGfKhJyftE30nI root@ubuntu
[ 18.469920] cloud-init[1160]: The key's randomart image is:
[ 18.470153] cloud-init[1160]: ++[EDC5519 256]++
[ 18.470356] cloud-init[1160]: |.+.B+|
[ 18.470687] cloud-init[1160]: |..+.Ec+|
[ 18.470902] cloud-init[1160]: |o o +.+=o|
[ 18.471100] cloud-init[1160]: |* + * +.+=o|
[ 18.471296] cloud-init[1160]: |B S + o.+|
[ 18.471493] cloud-init[1160]: |. o + o .|
[ 18.471761] cloud-init[1160]: |. = + . .|
[ 18.471937] cloud-init[1160]: |. o .|
[ 18.472102] cloud-init[1160]: |
[ 18.472263] cloud-init[1160]: +----[SHA256]-----+

ubuntu login: afifaarooj
Password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System Information as of Wed Sep 16 08:34:36 PM UTC 2020

System load:  0.0          Processes:      215
Usage of /:   50.5% of 9.75GB    Users logged in: 0
Memory usage: 21%          IPv4 address for ens33: 192.168.184.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

9 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

afifaarooj@ubuntu:~$
```

Terminal output verifying prompt format <github-username>@ubuntu:~\$.

Task 16: Find the Ubuntu Server IP Address

ubuntu_ip_addr_command.png

```
Usage of /:   50.5% of 9.75GB    Users logged in: 0
Memory usage: 21%          IPv4 address for ens33: 192.168.184.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

9 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

afifaarooj@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:10:0a:9d brd ff:ff:ff:ff:ff:ff
    altname enp00c2910ba9d
    inet 192.168.184.129/24 metric 100 brd 192.168.184.255 scope global dynamic ens33
        valid_lft 1162sec preferred_lft 1162sec
    inet6 fe80::c00c:2910:0a9d::/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
afifaarooj@ubuntu:~$
```

Terminal execution of ip addr command.

Ubuntu_ip_address.png

```
Usage of /: 50.5% of 9.75GB  Users logged in: 0
Memory usage: 21%  IPv4 address for ens33: 192.168.184.129
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

9 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

afifaarooj@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:10:b0:9d brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c2910b09d
    inet 192.168.184.129/24 metric 100 brd 192.168.184.255 scope global dynamic ens33
        valid_lft 1162sec preferred_lft 1162sec
    inet6 fe80::20c:29ff:fe10:b09d/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
afifaarooj@ubuntu:~$
```

Assigned IPv4 address highlighted on the primary network interface.

Task 17: Connect to Ubuntu Server from Windows

windows_ssh_command.png

```
afifaarooj@ubuntu: ~
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\afifa> ssh afifaarooj@192.168.184.129
afifaarooj@192.168.184.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro
```

Windows terminal executing `ssh <github-username>@<ubuntu-ip>`.

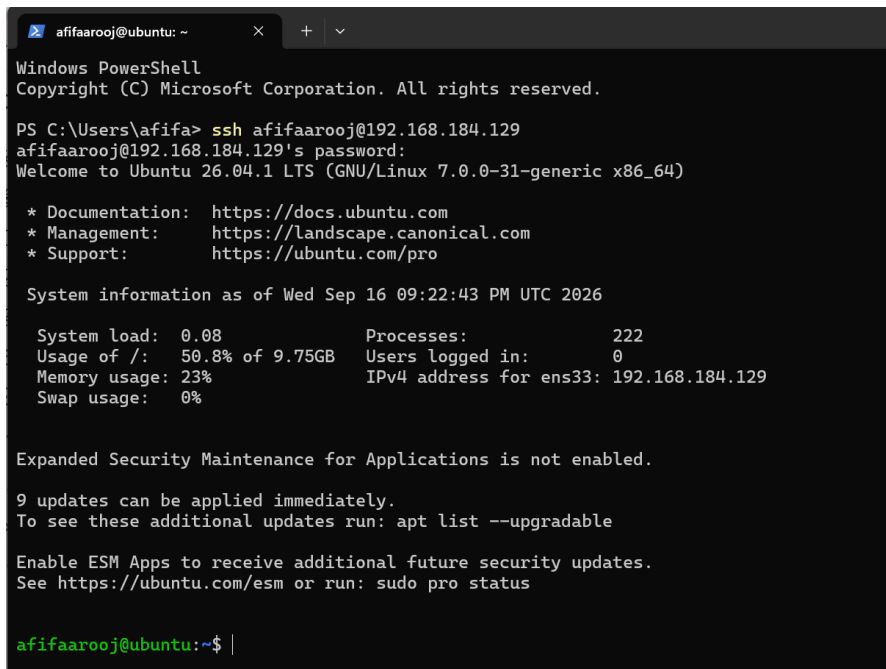
windows_ssh_fingerprint.png

```
afifaarooj@ubuntu: ~
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\afifa> ssh afifaarooj@192.168.184.129
The authenticity of host '192.168.184.129 (192.168.184.129)' can't be established.
ECDSA key fingerprint is SHA256:d5:b1:67:8e:b0:f3:b2:01:4d:5d:0f:71:34:77:90:
5a:2a:b1:3e:7a.
Are you sure you want to continue connecting (yes/no/fingerprint)]? yes
Warning: Permanently added '192.168.184.129' (ECDSA) to the list of known
hosts.
```

Host key fingerprint confirmation prompt accepted with 'yes'.

windows_ssh_login.png



```
afifaarooj@ubuntu: ~
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\afifa> ssh afifaarooj@192.168.184.129
afifaarooj@192.168.184.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Sep 16 09:22:43 PM UTC 2026

System load:  0.08               Processes:            222
Usage of /:   50.8% of 9.75GB    Users logged in:     0
Memory usage: 23%               IPv4 address for ens33: 192.168.184.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

9 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

afifaarooj@ubuntu:~$ |
```

Successful remote SSH authentication from Windows.

Windows_ssh_identity.png

```
afifaarooj@ubuntu: ~  
Windows PowerShell  
Copyright (C) Microsoft Corporation. All rights reserved.  
  
PS C:\Users\afifa> ssh afifaarooj@192.168.184.129  
afifaarooj@192.168.184.129's password:  
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Wed Sep 16 09:22:43 PM UTC 2026  
  
System load:  0.08      Processes:            222  
Usage of /:   50.8% of 9.75GB  Users logged in:     0  
Memory usage: 23%      IPv4 address for ens33: 192.168.184.129  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
9 updates can be applied immediately.  
To see these additional updates run: apt list --upgradable  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
afifaarooj@ubuntu:~$ |
```

Remote SSH prompt confirming session identity as `<github-username>@ubuntu`.

